ACADEMIC YEAR : 2026 - 2027

YEAR/SEM : IV/VII

AD1411 – PROJECT PHASE I REVIEW

BATCH NO : 01

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Guide Name : MRS.M.SARATHA

Name of the Candidate : AGALYA S

DIVYA SRI S

GOBIKA S

Register Number : 927623BAD003

927623BAD030

927623BAD035

- Keep each campus' s private training data within its institution.
- Support collaboration when campuses use different model architectures and non-IID data.
- Build a trust-weighted global teacher and adapt local knowledge distillation.

- Universities hold useful data, but sharing raw records may raise privacy and governance concerns.
- Federated Learning lets campuses train locally and collaborate without centralizing their datasets.
- Different data distributions and model architectures make direct model-update averaging difficult.
- Our project uses prediction-based knowledge distillation to share learning across campuses.

Year	Authors	Study / Paper	Method	Key benefit	Limitation
2019	McMahan et al.	Communication-Efficient Learning of Deep Networks from Decentralized Data	Federated Averaging (FedAvg)	Keeps training data on clients	Usually assumes compatible model parameters
2020	Li et al.	Federated Optimization in Heterogeneous Networks	FedProx	Addresses some client/data heterogeneity	Needs compatible model structures
2020	Li & Wang et al.	FedMD: Heterogeneous Federated Learning via Model Distillation	Prediction exchange + distillation	Allows different model architectures	Teacher quality depends on predictions
2021	Zhu et al.	Data-Free Knowledge Distillation for Heterogeneous Federated Learning	Federated knowledge distillation	Supports model heterogeneity	May need extra synthesis/communication

Year	Authors	Study / Paper	Method	Key benefit	Limitation
2018	Shokri & Shmatikov	Privacy-Preserving Deep Learning	Distributed training + privacy	Introduces privacy-aware collaboration	Privacy depends on protection settings
2017	Abadi et al.	Deep Learning with Differential Privacy	Differential Privacy (DP)	Formal privacy guarantees	Noise can reduce model utility
2020	Hinton et al. (concept)	Knowledge Distillation	Teacher– student learning	Transfers useful model knowledge	Teacher errors can transfer
Project	Our design	Adaptive Trust-Weighted Knowledge Distillation	Entropy trust + DP + adaptive KD	Reliability-aware collaboration	To be evaluated in Phase II

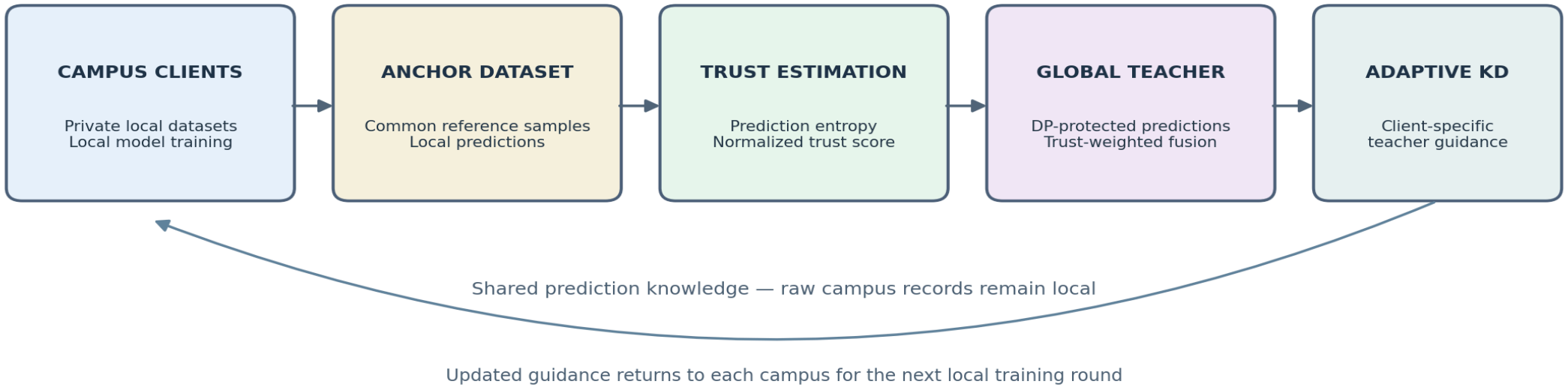
- Campus datasets are private and often have different distributions.
- Participating campuses may use different local model architectures.
- Equal contribution can give uncertain predictions too much influence.
- A fixed teacher–learning balance may not suit every campus.
- Need: privacy-aware collaboration that uses prediction reliability to guide learning.

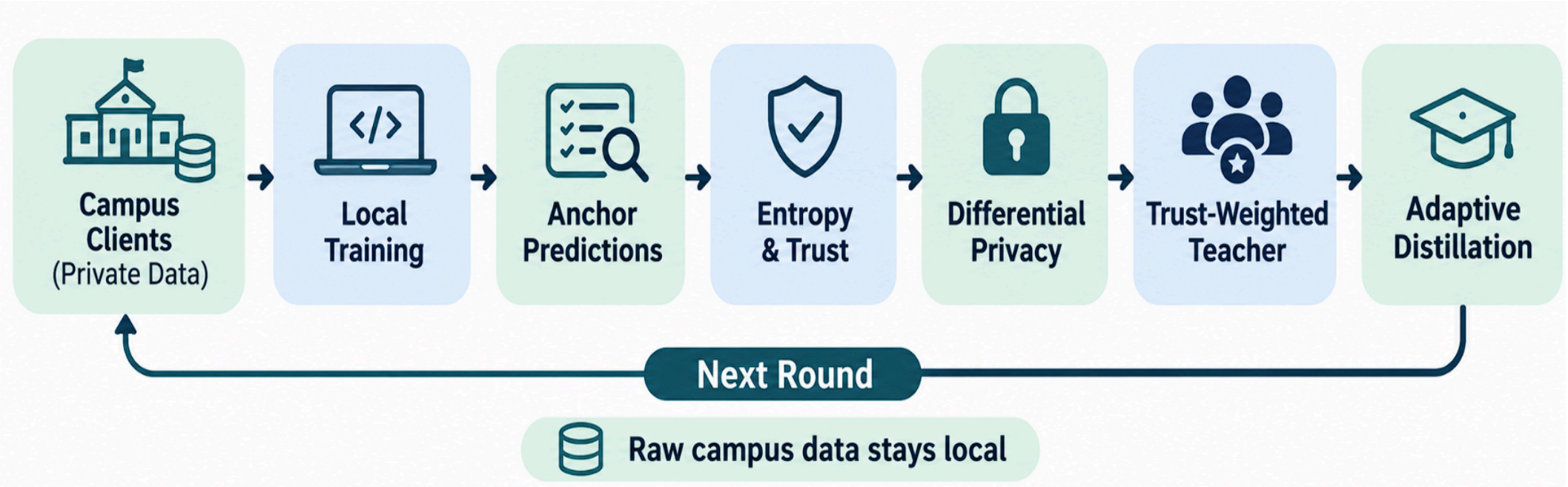
- Conventional federated methods often aggregate model parameters or updates.
- Parameter aggregation generally assumes compatible model structures.
- Some distillation approaches exchange predictions but do not adapt contribution using trust.
- Privacy risks may remain when prediction information is shared without protection.

Aspect	Common Existing Approaches	Proposed Design
Data location	Raw data remains local in FL	Raw data remains local
Model types	Parameter aggregation often needs compatible models	Prediction sharing supports heterogeneous models
Client contribution	Often equal or fixed weighting	Entropy-based trust weighting
Local learning	Fixed distillation settings in some methods	Trust-adaptive local / teacher balance
Privacy	Depends on method and shared information	DP protection applied to shared predictions
Technology Used	Traditional digital tools	Whisper, NLP, Stable Diffusion

- Each campus trains a model on its own private dataset.
- All campuses predict on a shared anchor dataset.
- Prediction entropy is converted into a trust score for each campus.
- Protected predictions are combined into a trust-weighted global teacher.
- Each campus updates its model using a trust-adaptive distillation objective.

ADAPTIVE TRUST-WEIGHTED KNOWLEDGE DISTILLATION




Campus Clients (Private Data)
Local Training
Anchor Predictions
Entropy & Trust
Differential Privacy
Trust-Weighted Teacher
Adaptive Distillation
Next Round
Raw campus data stays local

- Prediction-based collaboration supports different campus model architectures.
- Entropy-based trust estimates how reliable each campus' s anchor predictions are.
- Trust influences both global teacher construction and local learning balance.
- Differential Privacy is planned to reduce sensitive information inference from shared predictions.

- Combines entropy-based client trust with a trust-weighted global teacher.
- Uses each campus' s trust score to adapt local-versus-teacher learning.
- Designed for multi-campus collaboration with heterogeneous models and non-IID data.
- Phase I defines the design; experimental benefits will be verified in Phase II.

Software Requirements

- Python for implementation
- PyTorch for model training and knowledge distillation
- Flower framework for federated client coordination (planned)
- NumPy, Pandas, and Matplotlib for data handling and analysis
- Privacy mechanism: Differential Privacy for shared predictions
- Development tools: Jupyter Notebook or VS Code

- Phase I :

  Problem definition and motivation

  Literature survey and research gap

  System architecture and federated round workflow

  Core modules and experimental validation plan

- Phase II : implementation, experiments, comparison, and evaluation.

- Check local training on each campus dataset.
- Verify anchor predictions have consistent class dimensions.
- Test entropy and trust-score calculation.
- Check trust-weighted teacher aggregation and adaptive distillation.
- Evaluate privacy settings and utility trade-offs.
- Compare with baseline methods using planned metrics.

- Problem and research gap finalized.
- Literature review completed.
- Architecture and workflow defined.
- Modules and validation plan documented.
- Phase-II implementation is pending.

The proposed system, Adaptive Trust-Weighted Knowledge Distillation for Heterogeneous Multi-Campus Federated Learning, aims to improve collaborative learning among universities while preserving data privacy. It uses entropy-based trust estimation to identify reliable predictions, Differential Privacy to protect shared information, and a trust-weighted global teacher to guide local model training. Adaptive Knowledge Distillation enables collaboration between campuses with different data distributions and model architectures without sharing raw datasets. The proposed approach will be implemented and evaluated in Phase II to measure model performance, privacy, convergence, and communication cost.

THANK YOU